PRE-ALGEBRA (8TH) · UNIT 3: LINEAR AND FUNCTIONAL RELATIONSHIPS

Road Trip Rate Lab

Compare cars on a road trip using tables, graphs, and equations.

Honors

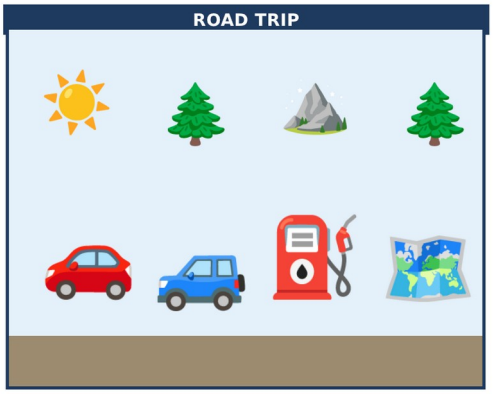
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your class is planning a road trip to a national park, and three family cars are going. Each car's speed is given a different way: one with a table, one with a graph, and one with an equation.

You will compare the cars, write functions for distance and fuel, and plan when everyone will arrive and how much the gas will cost.



Car A table

0 h → 0 mi · 1 h → 55 mi

2 h → 110 mi · 3 h → 165 mi

Car B graph

passes (1, 80) and (3, 200)

Car C

d = 62t

Learning Goals

- I can build linear functions from two points and interpret them in context.
- I can solve catch-up and comparison problems with linear functions.
- I can distinguish linear and nonlinear models and reason about average rates.

Project Timeline

When	What you will do
Day 1	Building Functions
Day 2	Chases and Curves
Day 3	Average Speed
Final	Road Trip Plan + reflection

Part 1: Building Functions

Day 1 · Use the car cards.

1. Car D was 130 miles from home after 2 hours and 310 miles after 5 hours. Write its function and interpret the initial value.

2. After how many hours is Car D 50 miles ahead of Car A (d = 55t)?

3. f(x) = 2x − 3 and g is given by the table (0, 1), (1, 4), (2, 7). Which has the greater rate of change? The greater initial value?

Part 2: Chases and Curves

Day 2 · Show your work.

4. Car C (62 mph) leaves 1.5 hours after Car A (55 mph) from the same place. When does Car C catch up, and how far have they gone?

5. Stopping distance is about $d = 0.05v^2$ feet at speed v mph. Find d at 20, 40, and 60 mph. Is the model linear?

6. Car A costs $0.12 per mile; Car E gets 24 mpg with gas at $3.36. Which costs more for 500 miles, and by how much?

Part 3: Average Speed

Day 3 · Plan a trip with several legs.

7. A car drives 2 hours at 60 mph and 1 hour at 30 mph. What is its average speed?

8. Explain why the answer is not (60 + 30) ÷ 2 = 45 mph.

9. Design a three-leg trip of 500 miles that arrives in exactly 9 hours. Give each leg's speed and time.

Final Product: Road Trip Plan

Your final product must include:

- Functions built from two points
- Catch-up and lead calculations
- Linear vs. nonlinear analysis
- Cost comparison of two cars
- A designed multi-leg trip with average speed

Reflection

R1. Why does a constant rate of change make a function linear?

R2. When is average speed equal to the average of the speeds?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Rates, functions, and comparisons are correct and interpreted in context. Goes beyond the minimum with a creative, realistic choice.	Rates, functions, and comparisons are correct and interpreted in context.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Trip plan shows each car's function, arrival time, and cost with clear labels. Polished, organized, and visually engaging.	Trip plan shows each car's function, arrival time, and cost with clear labels. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.